

BUYERS' GUIDE

ESTYLE

Breathe Easy With INTELLIGENT AIR PURIFIERS



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Air purifiers make use of two common types of filters to remove extremely small particles from the air. One is called high-efficiency particulate air (HEPA) filter, which uses very fine glass threads with diameter less than one micron to collect particles down to 0.3 micron in diameter.

Then there is the electrostatic precipitator (ESP) that uses electrostatic charge created by a cloud of free electrons, like plasma, into which dust particles are forced in. These filters collect and filter out particles down to a diameter of 0.01 micron (0.0001mm).

In addition, both types generally use special carbon filters to clear out volatile organic compounds and obnoxious smell due to burning fossil fuels, vehicle exhausts and chlorofluorocarbons.

The filters

Electrostatic precipitators draw polluted air into the purifier unit where the particles get charged using an ionising mechanism. Then, the particles pass through a filter that carries an opposite charge to that of the passing particles. The particles stick to the filter and are removed from flowing air.

Ion generators in some filters send out ions into the air instead of charging these inside the purification unit. When the surrounding air gets charged, ions attach themselves to airborne particles. In this process, these acquire charges and newly-

charged particles get attracted to surfaces of surrounding objects, walls and drapes, and fall or settle down on the floor with other particles.

Activated carbon filters. The activated carbon filters are used for odour reduction. These filter out volatile organic compounds and odour. Made up of carbon-impregnated cloth or foam, these are manufactured by infusing powdered activated carbon, which is wrapped around the inside or outside of the HEPA filter.

In an electrostatic precipitator unit, these are stretched in a frame at the inlet or outlet.

Pre-air filters. These help capture large dust particles that can clog the HEPA filter and, hence, are a must have.

What to look for

There are some elements that you should look at while shopping for an air purifier.

Performance. Performance and efficiency of air purifiers are measured by minimum efficiency reporting value (MERV) for air filters installed in the ductwork of HVAC systems. MERV performance value ranging from the one (lowest) to 20 (highest) could be used to compare the performance while buying these units.

MERV rating of one to four is commonly found in residential air-conditioning systems. Medium-efficiency filters fall in the range of five to 13 MERV, and these are capable of

removing small to large particles in the air. Filters having 14 to 20 MERV fall under high-efficiency range, and are used for large halls.

Portability. Air purifier manufacturers are also focussing on localised air cleaning, and are striving to make these handy and portable. The only flipside is that these smaller air purifiers may not perform as efficiently as the

Filter technology

